Math 242, S20 **Name:** Answer

Quiz 4

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (5 points) Find a general solution using the substitution method for the differential equation:

$$xy^2y' = 3x^3 + y^3.$$

(*Hint: First change it to the form of* $y' = f(y/x)$ *and then use the substitution* $v = y/x$)

$$y' = \frac{3x^3+y^3}{xy^2} = 3\frac{x^2}{y^2} + \frac{y}{x} = 3\left(\frac{x}{y}\right)^2 + \frac{y}{x}$$

let $v = \frac{y}{x}$, the $y = vx$ and $\frac{dy}{dx} = v + x\frac{dv}{dx}$

thus we have

$$v + x\frac{dv}{dx} = 3\left(\frac{1}{v}\right)^2 + v$$

$$x\frac{dv}{dx} = \frac{3}{v^2}$$

$$v^2 dv = \frac{3}{x}dx$$

$$\int v^2 dv = \int \frac{3}{x}dx$$

$$\frac{v^3}{3} = 3\ln|x| + C_1$$

$$v^3 = 9\ln|x| + C$$

$$\Rightarrow \left(\frac{y}{x}\right)^3 = 9\ln|x| + C$$

$$y^3 = x^3(9\ln|x| + C)$$

$$\underline{y = x(9\ln|x| + C)^{\frac{1}{3}}}$$

Problem 2. (5 points) Verify whether the given differential equation is **exact** and then solve it if yes:

$$(4x - y)\,dx + (6y - x)\,dy = 0.$$

$M(x,y) = 4x - y$, $N(x,y) = 6y - x$.

$$\left.\begin{array}{l} \dfrac{\partial M}{\partial y} = -1 \\[2ex] \dfrac{\partial N}{\partial x} = -1 \end{array}\right\} \Rightarrow \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \Rightarrow \text{exact!}$$

Now we solve for $F(x,y)$ such that $\frac{\partial F}{\partial x} = M$ & $\frac{\partial F}{\partial y} = N$.

$$\begin{aligned} F(x,y) &= \int M(x,y)\,dx + g(y) \\ &= \int 4x - y\,dx + g(y) \\ &= 2x^2 - xy + g(y) \end{aligned}$$

then we have.

$$\frac{\partial F}{\partial y} = \frac{\partial}{\partial y}[2x^2 - xy + g(y)] = N(x,y)$$

$$\Rightarrow -x + g'(y) = 6y - x$$

$$g'(y) = 6y \Rightarrow g(y) = 3y^2$$

$$\Rightarrow F(x,y) = 2x^2 - xy + 3y^2$$

So that the general solution of the equation is implicitly given by

$$2x^2 - xy + 3y^2 = C.$$